

Plane Geometry



Angle relationships

- 1 Two angles are supplementary, and one is 3 times the other. Find both angles.
 - 2 Two parallel lines are cut by a transversal, creating an angle of 70° . Find the measure of its alternate interior angle and its corresponding angle.
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Triangle angle sum and exterior angles

- 3 A triangle has angles measuring x , $2x$, and $3x$. Find each angle.
 - 4 A triangle's exterior angle at one vertex measures 115° . If one of the remote interior angles is 50° , find the other remote interior angle.
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Area and perimeter of polygons

- 5 Find the area of a triangle with base 14 and height 9.
 - 6 Find the area of a parallelogram with base 12 and height 5.
 - 7 Find the area of a regular hexagon with side length 6, using the formula $A = \frac{3\sqrt{3}}{2}s^2$.
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The Pythagorean theorem and special right triangles

- 8 A right triangle has legs 9 and 12. Find the hypotenuse.
 - 9 A 45-45-90 triangle has a leg of length 8. Find the length of the hypotenuse.
 - 10 A 30-60-90 triangle has a hypotenuse of 20. Find the lengths of both legs.
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Circles: area, circumference, arcs, and sectors

- 11 Find the area and circumference of a circle with diameter 14, in terms of π .
- 12 Find the arc length of a 120° sector in a circle with radius 9, in terms of π .

Similar and congruent figures

- 13 Two similar triangles have a scale factor of $\frac{3}{5}$. If the smaller triangle has area 18, find the area of the larger triangle.

Volume and surface area of 3D solids

- 14 Find the volume of a rectangular prism with dimensions 5, 8, and 10.
- 15 Find the volume of a cone with radius 6 and height 10, in terms of π .

Multi-step geometry problems

- 16 This question has two parts. A square garden has area 144 square feet. (a) Find the side length and perimeter of the garden. (b) A circular fountain with radius 3 feet is placed inside the garden. Find the remaining garden area (not covered by the fountain), in terms of π .

Trapezoids and composite polygon areas

- 17 Find the area of a trapezoid with parallel sides 8 and 14, and height 6.
- 18 An L-shaped figure is formed by a 10-by-8 rectangle with a 4-by-3 rectangle removed from one corner. Find the remaining area.

Chords, tangents, and inscribed angles in circles

- 19 An inscribed angle subtends an arc of 100° . Find the measure of the inscribed angle.
- 20 A tangent line touches a circle at point P . Explain the relationship between the tangent line and the radius drawn to point P .

Congruence and triangle proofs

- 21 State the triangle congruence criterion (SSS, SAS, ASA, AAS) that applies when two triangles share two equal angles and the included side.

A multi-part composite geometry problem

- 22** This question has two parts. A semicircular window has a diameter of 12 feet and sits atop a rectangular window of width 12 feet and height 5 feet. (a) Find the area of the semicircular portion, in terms of π . (b) Find the total area of the combined window shape.